
MedCore: Boundary-Preserving Medical Core Pruning for MedSAM

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Abstract

Medical segmentation foundation models such as SAM and MedSAM provide strong prompt-driven segmentation, but their image encoders are still too large for many clinical settings. Compression is also risky in medicine because a model can keep high Dice while losing boundary fidelity. We propose **MedCore**, a structured pruning framework for MedSAM. The main idea is to preserve two kinds of structures: structures that became important during SAM-to-MedSAM adaptation, and structures that have high boundary leverage. We identify the first type by a dual-intervention score that compares zeroing a group with resetting it to its original SAM weight. We identify the second type by boundary-aware Fisher estimation. We also introduce a boundary leverage principle, which shows that compression-induced boundary displacement is controlled by logit perturbation on the boundary divided by the logit spatial gradient. This principle explains why boundary metrics can degrade even when Dice remains high. On polyp segmentation benchmarks, MedCore reduces parameters by 60.0% and FLOPs by 58.4% while achieving Dice 0.9549, Boundary F1 0.6388, and HD95 5.14 after recovery fine-tuning. It also reaches **86.6%** parameter reduction and **90.4G** FLOPs with strong boundary quality. Our analysis further shows that MedSAM lies in a head-fragile boundary regime: head-pruning steps have 2.887 times larger 95th-percentile boundary leverage than MLP-pruning steps, and this logit-level effect is consistent with BF1 and HD95 degradation. Our code is available at <https://github.com/cenweizhang/MedCore>.

1 Introduction

Segment Anything Model (SAM) changed image segmentation by turning segmentation into a prompt-driven task [1]. MedSAM adapts this idea to medical images and gives strong results across many modalities [2]. These models are useful because they can accept prompts such as bounding boxes and produce high-quality masks. Their cost is still high. The MedSAM image encoder is a ViT-based model [3], and its inference cost can be difficult to support in real-time endoscopy, point-of-care ultrasound, and other resource-limited settings.

A natural solution is model compression. Yet medical segmentation has a failure mode that ordinary compression metrics can hide. A compressed model may still identify the coarse lesion region

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and keep a high Dice score, while its predicted boundary becomes thick, broken, or shifted. This is a serious issue because boundary quality affects lesion measurement, treatment planning, and clinical interpretation. This paper studies structured pruning for MedSAM under a boundary-sensitive setting, directly pruning the existing adapted model rather than training a lightweight surrogate. Since MedSAM is adapted from SAM instead of trained from scratch, we exploit the transition from the SAM checkpoint θ^S to the MedSAM checkpoint θ^M to identify which structures remain general-purpose and which become medical-specific.

We propose **MedCore**, a boundary-preserving medical core pruning framework that keeps a compact yet medically adapted MedSAM subnetwork. For each structure group g , MedCore uses a dual-intervention score that combines its current prediction contribution under zeroing and its medical-adaptation contribution under reset to the original SAM weights, which are two counterfactual questions. A group is safe to prune only when both signals suggest low importance.

We further introduce a boundary leverage principle by viewing the predicted mask boundary as a level set of the logit map. If pruning changes the logit map by $\delta_G(\mathbf{x})$, the first-order boundary displacement is proportional to $\delta_G(\mathbf{x})$ and inversely proportional to the spatial gradient norm $\|\nabla_{s_\theta}(\mathbf{x})\|_2$. This gives a direct geometric explanation of boundary collapse. Compression is dangerous when it creates a large logit perturbation on a boundary where the logit slope is small. We use this principle to motivate boundary-aware Fisher scoring and to analyze the different roles of attention heads and MLP connections.

In summary, Our three contributions are as follows: First, we formulate MedSAM pruning as medical-core extraction and introduce a dual-intervention score based on zeroing and reset-to-SAM operations. Second, we propose boundary-aware Fisher and Cross-Fisher approximations to make the intervention score practical and boundary-sensitive. Third, we introduce boundary leverage as a simple geometric principle for explaining Dice-boundary mismatch and head-MLP pruning trade-offs.

Our results further show that this distinction matters. Experiments on polyp endoscopy, breast ultrasound, and dermoscopy demonstrate that MedCore can remove a large fraction of MedSAM parameters while preserving strong region and boundary performance. In a 7×7 head-MLP pruning sweep, head-pruning steps exhibit $2.887 \times$ larger 95th-percentile boundary leverage than MLP-pruning steps, with all 15 valid paired comparisons favoring MLP pruning. At the metric level, the BF1 and HD95 damage densities of head pruning are respectively $2.607 \times$ and $2.432 \times$ those of MLP pruning, indicating that MedSAM lies in a head-fragile boundary regime: attention heads are not merely low-parameter components, but can have high boundary leverage.

2 Related Work

Promptable segmentation and medical SAM. SAM formulates segmentation as a promptable task using a ViT image encoder, a prompt encoder, and a lightweight mask decoder [1, 4]. MedSAM adapts this design to medical images via large-scale image-mask pairs [2]. Subsequent work improves medical adaptation along several axes: parameter-efficient tuning (SAMed, SAM-Med2D, Medical-SAM-Adapter [5, 6, 7]; Polyp-SAM for transfer to polyp segmentation [8]), boundary-aware feature design (I-MedSAM with implicit representations [9]; LDFSAM with distilled feature prompting [10]), 3D/video extension (MedSAM2 [11]), and feature fusion with task-specific networks [12, 13]. These methods focus on prompting, fusion, or boundary localization. We study a complementary question: how to *compress* an already adapted MedSAM checkpoint while preserving the structures that became important during SAM-to-MedSAM adaptation.

Efficient SAM and medical segmentation models. The heavy ViT image encoder of SAM has stimulated extensive research on efficient segmentation, including lightweight encoder replacement, knowledge distillation, quantization, and structured pruning [14, 15, 16, 17, 18]. Medical settings also have efficient segmentation backbones, including U-Net, nnU-Net, Swin-Unet, EMCAD, MK-UNet, and quantized medical SAM variants [19, 20, 21, 22, 23, 18]. Recent lightweight polyp models, such as HSSAM-Net, further show that careful multi-scale aggregation and boundary-aware enhancement can give strong accuracy with small parameter budgets [24]. These works confirm the need for efficient segmentation. Most of them redesign the architecture, train a compact surrogate, or quantize a model. MedCore instead directly prunes MedSAM and uses both the original SAM weights θ^S and the adapted MedSAM weights θ^M .

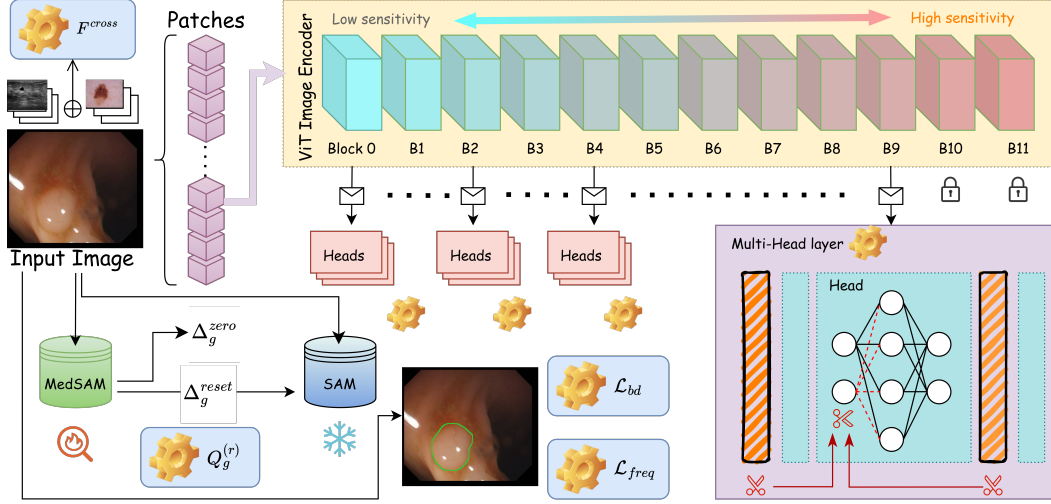


Figure 1: MedCore pruning overview. The Cross-Fisher signal F^{cross} combines SAM and MedSAM gradients to score attention heads and MLP channels via $Q_g^{(r)}$, driving block-sensitive cascade pruning of the ViT encoder (blocks B10–B11 protected) with boundary- and frequency-aware recovery. Yellow gears mark where pruning operations are applied: Cross-Fisher estimation, priority computation, per-block head pruning, and recovery-loss-driven fine-tuning.

Pruning and Transformer compression. Classic pruning methods estimate the loss increase caused by removing weights or groups. Optimal Brain Damage and Optimal Brain Surgeon use second-order information to approximate this increase [25, 26]. Deep Compression combines pruning, quantization, and coding [27]. Later methods use movement, Fisher information, Hessian-aware saliency, or group-level sensitivity to prune pretrained and Transformer models [28, 29, 30]. Attention-head pruning studies show that many heads can be redundant, but some specialized heads are important and task-dependent [31, 32]. This literature motivates structured pruning, but it usually estimates importance from one endpoint checkpoint.

Boundary preservation in medical segmentation. Medical segmentation models often optimize region losses such as Dice and cross-entropy. These losses can keep coarse foreground overlap but still miss fine contours. To address this limitation, prior work has explored boundary losses, surface- or Hausdorff-inspired objectives, topology-aware regularization, and boundary-aware architectures for more accurate contour modeling [33, 34, 35, 36, 37, 38, 39]. Different from these task-specific segmentation methods, our work brings boundary preservation into model compression by asking not only whether pruning preserves Dice, but also whether it perturbs boundary-critical structures and the boundary logit field.

3 MedCore: Boundary-Preserving Medical Core Pruning

MedCore has two design goals. The first goal is to preserve structures that became important during medical adaptation. The second goal is to preserve structures whose removal causes large boundary displacement.

3.1 Problem Formulation

A training or calibration sample is written as $z = (I, \mathbf{b}, M)$, where $I \in \mathbb{R}^{3 \times H \times W}$ is a medical image, $\mathbf{b} \in \mathbb{R}^4$ is a bounding-box prompt, and $M \in \{0, 1\}^{H \times W}$ is a binary mask. MedSAM maps $(I, \mathbf{b})$ to a logit map $s_{\theta, \phi}(I, \mathbf{b})$ and a probability mask $\hat{M} = \sigma(s_{\theta, \phi}(I, \mathbf{b}))$. The parameter vector $\theta = (\theta_E, \theta_D)$ contains the image encoder and mask decoder. The prompt encoder parameter ϕ is frozen.

Given a calibration set $\mathcal{D}_{\text{cal}} = \{\mathbf{z}_n\}_{n=1}^N$, we define the empirical medical risk as

$$\widehat{\mathcal{R}}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{n=1}^N \ell_{\text{seg}}(\mathbf{z}_n; \boldsymbol{\theta}), \quad \ell_{\text{seg}} = \ell_{\text{Dice}} + \ell_{\text{BCE}}. \quad (1)$$

We partition pruneable parameters into structured groups $\mathcal{G} = \{g_1, \dots, g_K\}$. In our implementation, groups are attention heads and MLP connection groups in the ViT image encoder. A binary group mask $\mathbf{s} \in \{0, 1\}^K$ gives the masked parameter $\mathbf{s} \odot_{\mathcal{G}} \boldsymbol{\theta}$. If c_g is the cost of group g , the ideal pruning problem is

$$\min_{\boldsymbol{\theta}, \mathbf{s}} \widehat{\mathcal{R}}(\mathbf{s} \odot_{\mathcal{G}} \boldsymbol{\theta}) \quad \text{s.t.} \quad \sum_{g \in \mathcal{G}} s_g c_g \leq B. \quad (2)$$

This problem is combinatorial. MedCore approximates it by scoring groups and removing low-score groups under a block-aware budget. We keep two checkpoints: $\boldsymbol{\theta}^S$ is the original SAM checkpoint, and $\boldsymbol{\theta}^M$ is the MedSAM checkpoint. Pruning starts from $\boldsymbol{\theta}^M$, while $\boldsymbol{\theta}^S$ serves as the adaptation reference.

3.2 Boundary Leverage Principle

We now state the geometric principle that motivates boundary-aware pruning. For a fixed image and prompt, we write the logit map as $s_{\boldsymbol{\theta}}(\mathbf{x})$, where $\mathbf{x}$ indexes a spatial location. The predicted boundary is the zero level set of this logit map. A compression operation G changes the model to $\boldsymbol{\theta}^{-G}$ and induces the logit perturbation $\delta_G(\mathbf{x}) = s_{\boldsymbol{\theta}^{-G}}(\mathbf{x}) - s_{\boldsymbol{\theta}}(\mathbf{x})$. The following assumption is standard in level-set perturbation analysis. It only says that the boundary is not locally flat in the logit field.

Assumption 3.1 (Regular boundary). For a given image and prompt, the logit map $s_{\boldsymbol{\theta}}$ is continuously differentiable in a neighborhood of its zero level set $\Gamma_{\boldsymbol{\theta}} = \{\mathbf{x} : s_{\boldsymbol{\theta}}(\mathbf{x}) = 0\}$. There exists $\kappa > 0$ such that $\|\nabla s_{\boldsymbol{\theta}}(\mathbf{x})\|_2 \geq \kappa$ for every $\mathbf{x} \in \Gamma_{\boldsymbol{\theta}}$.

Definition 3.2 (Boundary leverage). Let G be a compression operation with compression gain $\Delta C_G > 0$. Let $\mathcal{B}$ be a narrow boundary band around a ground-truth or predicted boundary. We define the boundary leverage of G as

$$\lambda_G^{\text{bd}} = \frac{1}{\Delta C_G} \mathbb{E}_{\mathbf{x} \in \mathcal{B}} \left[\frac{|\delta_G(\mathbf{x})|}{\|\nabla s_{\boldsymbol{\theta}}(\mathbf{x})\|_2 + \varepsilon} \right]. \quad (3)$$

Here $\varepsilon > 0$ is a small constant. This quantity estimates the boundary displacement caused by one unit of compression.

Theorem 3.3 (First-order boundary perturbation). Under Assumption 3.1, suppose that δ_G is sufficiently small in a neighborhood of $\Gamma_{\boldsymbol{\theta}}$. For each $\mathbf{x} \in \Gamma_{\boldsymbol{\theta}}$, let $u_G(\mathbf{x})$ be the signed normal displacement from the old boundary to the new boundary induced by G . Then

$$u_G(\mathbf{x}) = -\frac{\delta_G(\mathbf{x})}{\|\nabla s_{\boldsymbol{\theta}}(\mathbf{x})\|_2} + O(\|\delta_G\|_{C^1}^2). \quad (4)$$

Theorem 3.3 explains why boundary metrics can fail before Dice fails. Boundary displacement depends on the logit perturbation at the boundary and on the local logit slope. A group with small parameter count can still be dangerous if it causes a large boundary logit perturbation. We prove the theorem in Appendix A.

Proposition 3.4 (Boundary-aware budget rule). Let c_H and c_M be compression budgets assigned to heads and MLPs, and let $\mathcal{E}_{\text{bd}}(c_H, c_M)$ be a locally differentiable boundary error. If

$$\frac{\partial \mathcal{E}_{\text{bd}}}{\partial c_H} > \frac{\partial \mathcal{E}_{\text{bd}}}{\partial c_M}$$

at a feasible interior allocation, then moving a small amount of compression budget from head pruning to MLP pruning decreases boundary error to first order while keeping the total compression fixed.

Proposition 3.4 turns boundary leverage into a budget principle. It does not say that heads are always more important than MLPs. It says that the safer compression direction is the one with lower marginal boundary damage. Section 5.2 verifies that, for MedSAM, head-pruning steps have much higher boundary leverage than MLP-pruning steps.

3.3 Medical-Core Scoring via Dual Intervention

The boundary leverage principle says which structures are dangerous for boundaries. We also need to preserve structures that became important during medical adaptation. MedSAM has a useful reference checkpoint, namely the original SAM checkpoint θ^S . We therefore score each group g by two counterfactual interventions.

The zero intervention sets group g to zero while keeping all other groups unchanged. If T_g^0 denotes this operation, the exact zero cost is

$$\Delta_g^{\text{zero}} = \widehat{\mathcal{R}}(s \odot_{\mathcal{G}} T_g^0(\theta^M)) - \widehat{\mathcal{R}}(s \odot_{\mathcal{G}} \theta^M). \quad (5)$$

A small value means that g contributes little to current MedSAM predictions.

The reset intervention replaces group g in MedSAM by its SAM value. If T_g^S denotes this operation, the exact reset cost is

$$\Delta_g^{\text{reset}} = \widehat{\mathcal{R}}(s \odot_{\mathcal{G}} T_g^S(\theta^M)) - \widehat{\mathcal{R}}(s \odot_{\mathcal{G}} \theta^M). \quad (6)$$

A large reset cost means that the change from θ_g^S to θ_g^M is functionally important for medical segmentation. This signal is different from the zero cost. It measures adaptation-specific importance rather than only endpoint importance. We combine both costs as

$$Q_g = \alpha_{b(g)} \widehat{\Delta}_g^{\text{zero}} + (1 - \alpha_{b(g)}) \widehat{\Delta}_g^{\text{reset}}, \quad (7)$$

where $b(g)$ is the Transformer block containing g . We use a block-level mixture weight because different blocks respond differently to boundary and adaptation signals. The pruning priority is

$$P_g = \frac{Q_g}{(c_g + \varepsilon)^\tau}, \quad (8)$$

where $\tau \geq 0$ controls cost normalization. Groups with smaller P_g are pruned first.

3.4 Boundary-Aware Fisher and Cross-Fisher Approximation

Computing Eq. (5) and Eq. (6) exactly for every group is expensive. We use a local second-order approximation. Around θ^M , zeroing group g corresponds to the perturbation $-\theta_g^M$, and resetting group g corresponds to the perturbation $\theta_g^S - \theta_g^M$. We approximate the Hessian by diagonal Fisher information.

The loss used for Fisher estimation should reflect the metric we want to preserve. We therefore use a boundary-weighted loss. Let $B(M)$ be a morphological boundary map computed by dilation minus erosion. With pixel index u , we define

$$\ell_{\text{bd}}(z; \theta) = \sum_u (1 + \lambda_{\text{bd}} B(M)_u) \ell_{\text{BCE}}(s_{\theta, \phi}(I, \mathbf{b})_u, M_u) + \ell_{\text{Dice}}(\widehat{M}, M). \quad (9)$$

For a sub-distribution r , such as one dataset or modality, the MedSAM Fisher is

$$F_{r,i}^M = \frac{1}{|\mathcal{D}_{\text{cal}}^{(r)}|} \sum_{\mathbf{z}_n \in \mathcal{D}_{\text{cal}}^{(r)}} \left(\frac{\partial \ell_{\text{bd}}(\mathbf{z}_n; \theta^M)}{\partial \theta_i} \right)^2. \quad (10)$$

This gives the zero approximation

$$\widehat{\Delta}_g^{\text{zero},(r)} = \frac{1}{2} \sum_{i \in g} F_{r,i}^M (\theta_i^M)^2.$$

For the reset cost, the perturbation is the SAM-to-MedSAM parameter change. We also compute the SAM Fisher F_i^S on the same calibration data and use a geometric mean,

$$F_{r,i}^{\text{cross}} = \sqrt{F_{r,i}^M F_i^S + \varepsilon_F}. \quad (11)$$

The reset approximation becomes

$$\widehat{\Delta}_g^{\text{reset},(r)} = \frac{1}{2} \sum_{i \in g} F_{r,i}^{\text{cross}} (\theta_i^M - \theta_i^S)^2. \quad (12)$$

The geometric mean down-weights parameters that are insensitive in either SAM or MedSAM. This prevents the reset score from being dominated by large but functionally irrelevant weight shifts.

3.5 Distribution-Aware Aggregation and Budget Allocation

Medical data are heterogeneous. A structure that appears redundant on one dataset may be important on another dataset. We compute per-distribution scores $\hat{Q}_g^{(r)}$ using Eq. (7) and aggregate them as

$$\hat{Q}_g^{\text{dist}} = \sum_{r=1}^R \pi_r \hat{Q}_g^{(r)} + \beta \text{Var}_r[\hat{Q}_g^{(r)}], \quad (13)$$

where $\pi_r \geq 0$ and $\sum_r \pi_r = 1$. The variance term makes pruning more conservative for groups that are important on only some sub-distributions.

We also allocate pruning budgets non-uniformly across Transformer blocks. Let $\mathcal{B}_\ell$ be the parameter set of block ℓ . We estimate block sensitivity by

$$S_\ell = \sum_{i \in \mathcal{B}_\ell} F_i, \quad F_i = \sum_r \pi_r F_{r,i}^M.$$

Blocks with high S_ℓ receive smaller pruning quotas. In the main one-time pruning experiments, the deepest blocks are protected because they are closest to the mask decoder and have the highest Fisher sensitivity. In the extreme sequential setting, we apply a second conservative pruning pass to previously protected blocks after the first checkpoint has been selected.

3.6 Head-to-MLP Cascade and Recovery Fine-Tuning

MedCore prunes in a cascade. We first prune attention heads. Then we run a short recovery phase. We then prune MLP connection groups from the head-pruned model. This schedule avoids removing two different functional families at the same time. It also lets us analyze the marginal boundary effect of head and MLP pruning separately.

After pruning, we recover the remaining weights with a lightweight objective:

$$\mathcal{L}_{\text{rec}} = \mathcal{L}_{\text{seg}} + \lambda_1 \mathcal{L}_{\text{bd}} + \lambda_2 \mathcal{L}_{\text{feat}} + \lambda_3 \mathcal{L}_{\text{logit}} + \lambda_4 \mathcal{L}_{\text{freq}}. \quad (14)$$

Here $\mathcal{L}_{\text{bd}}$ is boundary-weighted BCE, $\mathcal{L}_{\text{feat}}$ distills encoder features from the unpruned MedSAM teacher, $\mathcal{L}_{\text{logit}}$ distills boundary-region logits, and $\mathcal{L}_{\text{freq}}$ penalizes high-frequency mask discrepancies. Recovery is not used to define the pruning score. It lets the remaining structures re-coordinate after structural removal.

4 Experiments

4.1 Experimental Setup

We evaluate MedCore on MedSAM with a ViT-B image encoder. The base model is the official MedSAM checkpoint, and the reference checkpoint is SAM ViT-B. We freeze the prompt encoder in all experiments. We use five medical image segmentation datasets across three modalities: CVC-ClinicDB [40], CVC-ColonDB [41], and Kvasir-SEG [42] for polyp endoscopy, BUSI [43] for breast ultrasound, and ISIC2018 [44, 45] for dermoscopy. Each dataset is split into training and test sets with a fixed random seed. For Fisher estimation, we sample 128 calibration images per dataset. We use batch size 1 for Fisher estimation to preserve per-sample gradients.

We report Dice, Intersection over Union (IoU), Boundary F1 (BF1), and 95th-percentile Hausdorff distance (HD95). We also report parameters and FLOPs. BF1 and HD95 are the key metrics for boundary fidelity. Unless stated otherwise, the main MedCore configurations include post-pruning recovery fine-tuning. The component ablation in Section 5.1 is reported without post-pruning fine-tuning for the one-time settings, so that it isolates the direct effect of each pruning component.

4.2 Main Results on Polyp Segmentation

Table 1 reports macro-averaged results on the three polyp benchmarks. MedCore reaches strong compression while preserving boundary quality after recovery. The h50_m70 configuration removes 60.0% of parameters and reduces FLOPs by 58.4%, while achieving Dice 0.9549, BF1 0.6388, and

Table 1: Macro-averaged results on CVC-ClinicDB [40], CVC-ColonDB [41], and Kvasir-SEG [42]. MedCore configurations include post-pruning fine-tuning. One-time uses a single head-to-MLP cascade. Sequential starts from h70_m95 and applies a second conservative pruning pass.

Category	Method	Params↓	FLOPs↓	Dice↑	IoU↑	BF1↑	HD95↓
Baseline	MedSAM [2]	89.7M	926.5G	0.9191	0.8648	0.5321	21.29
Efficient SAM	EfficientSAM-s [15]	26.4M	188.0G	0.8765	0.8131	0.4340	29.73
	EfficientSAM-t [15]	10.2M	55.9G	0.8723	0.8103	0.4218	25.98
	SlimSAM [17]	28.0M	189.9G	0.8435	0.7758	0.3922	34.11
Medical Seg.	Swin-Unet [21]	27.2M	5.92G	0.7845	0.6933	0.4347	23.88
	nnU-Net [19, 20]	46.3M	75.1G	0.9093	0.8574	0.5382	32.15
	EMCAD [22]	26.8M	10.6G	0.9141	0.8608	0.5305	30.31
	MK-UNet [23]	0.32M	0.61G	0.8658	0.7986	0.4635	46.21
Medical SAM	SAMed [5]	91.2M	206.7G	0.7419	0.6640	0.3283	97.57
	QMedSAM [18]	9.79M	75.7G	0.8089	0.7108	0.4118	14.55
MedCore One-time	h40_m30	61.3M	620.7G	0.9547	0.9173	0.6508	7.23
	h50_m70	35.8M	385.2G	0.9549	0.9169	0.6388	5.14
	h70_m70	30.1M	287.5G	0.9524	0.9130	0.6342	6.23
	h70_m95	24.9M	245.0G	0.9522	0.9124	0.6259	6.78
MedCore Sequential	h60_m70	15.6M	132.6G	0.9547	0.9176	0.6506	5.91
	h72_m84	13.7M	112.4G	0.9536	0.9153	0.6337	5.47
	h84_m95	12.1M	90.4G	0.9550	0.9174	0.6462	5.12

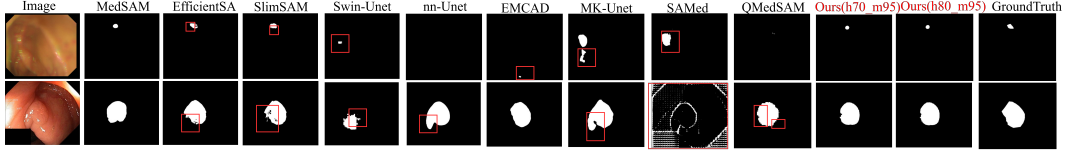


Figure 2: Qualitative segmentation comparison between our two high-pruning configurations and other models. Incorrect segmentation regions and boundary artifacts are highlighted with red boxes.

HD95 5.14. The h70_m95 configuration removes 72.2% of parameters and still obtains BF1 0.6259 and HD95 6.78. The sequential configuration h84_m95 reaches 12.1M parameters and 90.4G FLOPs, while keeping strong region and boundary metrics.

We present these results as compressed models after the same recovery pipeline. The improvement over the original MedSAM checkpoint should be read together with this recovery setting. The main conclusion is that MedCore keeps the benefit of boundary-aware recovery under heavy compression, rather than merely keeping region-level overlap.

4.3 Cross-Modality Generalization

Table 2 evaluates the same pruning pipeline on CVC-ClinicDB, BUSI, and ISIC2018. On ClinicDB and ISIC2018, MedCore improves boundary metrics after recovery. On BUSI, moderate compression preserves performance closely, while aggressive compression begins to degrade BF1 and HD95. This pattern is useful because it shows both the strength and the limit of direct cross-modality transfer. MedCore is not a replacement for target-domain validation. It is a compression framework that can preserve strong performance when the calibration and recovery data are representative.

5 Analysis and Ablation

5.1 Component Ablation: What Drives Boundary Preservation?

Table 3 ablates the main components. The one-time columns do not use post-pruning fine-tuning, so they isolate the structural effect of pruning decisions. The sequential column uses the same recovery protocol for all variants.

Table 2: Cross-modality results with post-pruning fine-tuning.

Dataset	Method	Params↓	FLOPs↓	Dice↑	BF1↑	HD95↓
CVC-ClinicDB [40]	MedSAM	89.7M	926.5G	0.9264	0.5477	13.77
	h50_m70	35.8M	377.0G	0.9601	0.6603	2.64
	h70_m70	30.1M	283.4G	0.9564	0.6409	3.85
	h70_m95	24.9M	240.9G	0.9523	0.6194	6.30
	h80_m95	23.2M	224.6G	0.9507	0.5951	4.46
BUSI [43]	MedSAM	89.7M	926.5G	0.9334	0.4989	7.30
	h50_m70	35.8M	377.0G	0.9357	0.5048	7.35
	h70_m70	30.1M	283.4G	0.9336	0.4913	7.82
	h70_m95	24.9M	240.9G	0.9298	0.4703	9.43
	h80_m95	23.2M	224.6G	0.9316	0.4833	8.58
ISIC2018 [44, 45]	MedSAM	89.7M	926.5G	0.9314	0.2401	7.47
	h50_m70	35.8M	377.0G	0.9525	0.4806	6.12
	h70_m70	30.1M	283.4G	0.9525	0.4732	5.57
	h70_m95	24.9M	240.9G	0.9530	0.4753	5.86
	h80_m95	23.2M	224.6G	0.9526	0.4708	5.94

Table 3: Component ablation on macro-average BF1 across the three polyp benchmarks. One-time columns are reported without post-pruning fine-tuning. The sequential column uses the same post-pruning fine-tuning for all variants. Δ is the difference from the full method at the same compression level.

Method	h40_m30		h50_m70		h70_m70		h70_m95		Seq. h84_m95	
	BF1	Δ	BF1	Δ	BF1	Δ	BF1	Δ	BF1	Δ
Full MedCore	0.5162	–	0.4078	–	0.3144	–	0.4050	–	0.6462	–
w/o Boundary Fisher	0.5035	-0.013	0.3724	-0.035	0.3096	-0.005	0.2781	-0.127	0.6304	-0.016
w/o reset	0.5149	-0.001	0.3967	-0.011	0.3628	+0.048	0.3227	-0.082	0.6279	-0.018
w/o variance	0.5221	+0.006	0.3997	-0.008	0.3812	+0.067	0.3249	-0.080	0.6462	0.000

Boundary-aware Fisher is the most direct boundary-preserving component. Removing it hurts BF1, and the effect is strongest at the most aggressive one-time setting h70_m95. The reset-to-SAM score has a different pattern. It has a small effect at moderate compression, but it becomes important when the pruning budget is severe. This supports our interpretation of reset scoring as an aggressive-compression safeguard. The distribution variance term follows a similar pattern. It protects structures that are unstable across datasets, and this becomes more useful when few structures remain.

The ablation suggests a hierarchy. Boundary Fisher is the strongest component for direct contour protection. Reset-to-SAM and variance aggregation mainly act as safeguards under aggressive compression. This hierarchy is consistent with the design of MedCore. We use boundary Fisher to protect boundary-core structures, and we use reset-to-SAM to protect medical-core structures that were created by adaptation.

5.2 Boundary Leverage Reveals a Head-Fragile Regime

We test the empirical prediction in Definition 3.2 using a 7×7 sweep over $h \in \{0.3, \dots, 0.9\}$ and $m \in \{0.3, 0.5, 0.7, 0.8, 0.85, 0.9, 0.95\}$, with 384 calibration images (128 each from Kvasir, ClinicDB, ColonDB). For each adjacent head step $G_H(i, j): (h_i, m_j) \rightarrow (h_{i+1}, m_j)$ and MLP step $G_M(i, j): (h_i, m_j) \rightarrow (h_i, m_{j+1})$, we compute the 95th-percentile version of Eq. (3) on the boundary band, skipping steps with zero parameter reduction.

Table 4 shows that head-pruning steps have substantially larger boundary leverage: median λ_{95}^{bd} is 3.961 for heads versus 1.372 for MLPs (ratio 2.887), and head wins in all 15 valid paired comparisons. This supports the empirical law $\text{Median}(\lambda_G^{\text{bd}} \mid G \in \mathcal{H}) > \text{Median}(\lambda_G^{\text{bd}} \mid G \in \mathcal{M})$ for MedSAM.

The BF1 and HD95 rows confirm at the final-mask level that logit-level boundary leverage translates to actual metric damage: head pruning incurs $2.4\text{--}2.6\times$ greater BF1 and HD95 degradation per one percent parameter reduction. We do not claim heads affect *only* boundary pixels—their full-image

Table 4: Step-level boundary leverage analysis on the head-MLP pruning sweep. The logit-level boundary leverage result agrees with metric-level BF1 and HD95 damage densities.

Quantity	Head steps	MLP steps	Head/MLP ratio
Median λ_{95}^{bd}	3.961	1.372	2.887
BF1 damage density	0.007692	0.002951	2.607
HD95 damage density	0.025406	0.010445	2.432
Valid steps	35	21	—
Paired win rate	15/15 = 1.000		

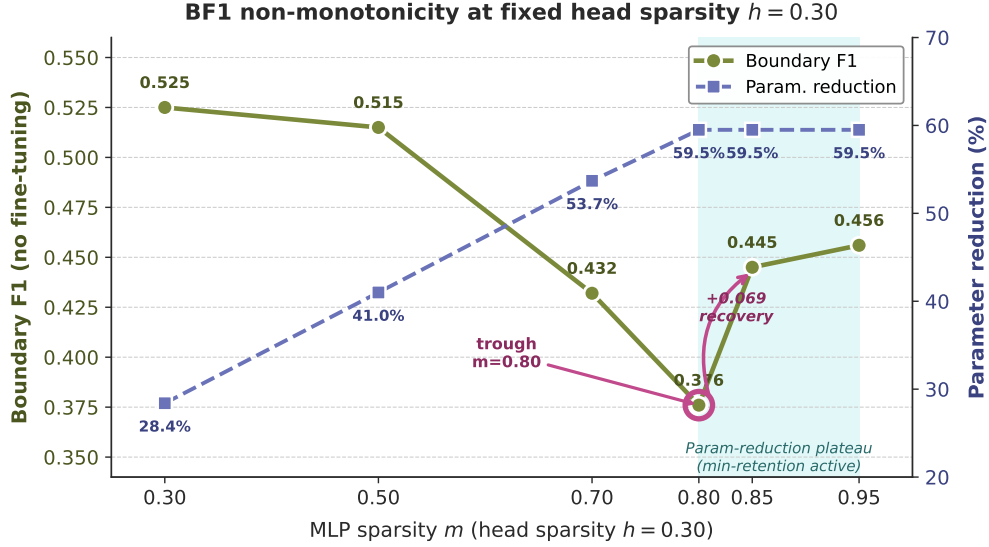


Figure 3: BF1 non-monotonicity at fixed head sparsity $h = 0.30$ (no fine-tuning). BF1 drops to a trough of 0.376 at $m = 0.80$, then recovers to 0.456 at $m = 0.95$ along the 59.5% parameter-reduction plateau where the min-retention constraint is active.

perturbation ratio is also high. Our narrower claim is that MedSAM heads are high-boundary-leverage components, making aggressive head pruning risky for boundary-sensitive segmentation.

5.3 Head-MLP Compression Landscape

The boundary leverage result above explains a non-trivial compression landscape: equal parameter reductions can produce markedly different BF1 depending on the head/MLP allocation. Near 60% reduction, h30_m80 yields BF1 0.376 while h50_m70 yields 0.408 (no fine-tuning)—parameter count alone does not determine boundary quality. Figure 3 shows one slice at fixed $h = 0.30$: BF1 first drops as m increases, then recovers once the effective parameter count saturates at the min-retention plateau. This non-monotonic behavior confirms that compression is not a scalar sparsity problem—both block allocation and structure type matter.

This result does not mean that extreme sparsity is always better. It means that boundary-aware allocation can matter as much as the nominal sparsity level. Together with Table 4, it supports a simple practical rule: head pruning should be conservative, and most of the compression should come from lower-leverage MLP structures unless a calibration analysis shows otherwise.

6 Conclusion

We introduced MedCore, a structured pruning framework for MedSAM. MedCore treats pruning as medical-core extraction. It preserves structures that are important after SAM-to-MedSAM adaptation through a reset-to-SAM intervention, and it preserves contour-critical structures through boundary-

aware Fisher estimation. We also introduced boundary leverage as a geometric view of compression-induced boundary damage. This view explains why a compressed MedSAM can keep high Dice while losing boundary fidelity, and it explains why attention heads require conservative pruning in our setting. Experiments show that MedCore can greatly reduce parameter count and FLOPs while keeping strong region and boundary performance across medical segmentation datasets. We note that MedCore inherits MedSAM’s underlying capabilities and biases; clinical deployment therefore requires task-specific validation and prospective safety evaluation.

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Appendix

The appendix contains proofs, implementation details, and additional information about the boundary leverage analysis. We keep the main text focused on the method and key empirical findings.

A Proofs

A.1 Proof of Theorem 3.3

Proof. Fix a point $\mathbf{x} \in \Gamma_\theta$. By definition of the old boundary, $s_\theta(\mathbf{x}) = 0$. Since Assumption 3.1 gives $\|\nabla s_\theta(\mathbf{x})\|_2 > 0$, the unit normal direction of the level set at $\mathbf{x}$ is

$$\mathbf{n}(\mathbf{x}) = \frac{\nabla s_\theta(\mathbf{x})}{\|\nabla s_\theta(\mathbf{x})\|_2}.$$

The new logit map after compression is $\tilde{s}(\mathbf{x}) = s_\theta(\mathbf{x}) + \delta_G(\mathbf{x})$. We seek the new boundary point on the normal line through $\mathbf{x}$. We write it as

$$\tilde{\mathbf{x}} = \mathbf{x} + u_G(\mathbf{x})\mathbf{n}(\mathbf{x}),$$

where $u_G(\mathbf{x})$ is the signed normal displacement. Because $\tilde{\mathbf{x}}$ lies on the new boundary, it satisfies

$$0 = \tilde{s}(\tilde{\mathbf{x}}) = s_\theta(\tilde{\mathbf{x}}) + \delta_G(\tilde{\mathbf{x}}).$$

We expand both terms around $\mathbf{x}$. The first-order Taylor expansion gives

$$s_\theta(\tilde{\mathbf{x}}) = s_\theta(\mathbf{x}) + u_G(\mathbf{x})\nabla s_\theta(\mathbf{x})^\top \mathbf{n}(\mathbf{x}) + O(u_G(\mathbf{x})^2).$$

Since $s_\theta(\mathbf{x}) = 0$ and $\nabla s_\theta(\mathbf{x})^\top \mathbf{n}(\mathbf{x}) = \|\nabla s_\theta(\mathbf{x})\|_2$, this becomes

$$s_\theta(\tilde{\mathbf{x}}) = u_G(\mathbf{x})\|\nabla s_\theta(\mathbf{x})\|_2 + O(u_G(\mathbf{x})^2).$$

The perturbation term satisfies

$$\delta_G(\tilde{\mathbf{x}}) = \delta_G(\mathbf{x}) + O(|u_G(\mathbf{x})|\|\delta_G\|_{C^1}).$$

Substituting the two expansions into the boundary equation gives

$$0 = u_G(\mathbf{x})\|\nabla s_\theta(\mathbf{x})\|_2 + \delta_G(\mathbf{x}) + O(u_G(\mathbf{x})^2) + O(|u_G(\mathbf{x})|\|\delta_G\|_{C^1}).$$

For small perturbations, the implicit function theorem guarantees that $u_G(\mathbf{x})$ is also small and has the same order as $\delta_G(\mathbf{x})$. Therefore the two remainder terms are second order in $\|\delta_G\|_{C^1}$. Solving the first-order part gives

$$u_G(\mathbf{x}) = -\frac{\delta_G(\mathbf{x})}{\|\nabla s_\theta(\mathbf{x})\|_2} + O(\|\delta_G\|_{C^1}^2).$$

This is Eq. (4). □

A.2 Proof of Proposition 3.4

Proof. Let the current allocation be (c_H, c_M) and keep the total compression fixed at $C = c_H + c_M$. We move a small amount $\eta > 0$ of compression budget from heads to MLPs. The new allocation is

$$(c_H - \eta, c_M + \eta).$$

The first-order Taylor expansion of $\mathcal{E}_{\text{bd}}$ at (c_H, c_M) gives

$$\mathcal{E}_{\text{bd}}(c_H - \eta, c_M + \eta) = \mathcal{E}_{\text{bd}}(c_H, c_M) - \eta \frac{\partial \mathcal{E}_{\text{bd}}}{\partial c_H} + \eta \frac{\partial \mathcal{E}_{\text{bd}}}{\partial c_M} + o(\eta).$$

The first-order change in boundary error is therefore

$$\Delta \mathcal{E}_{\text{bd}} = \eta \left(\frac{\partial \mathcal{E}_{\text{bd}}}{\partial c_M} - \frac{\partial \mathcal{E}_{\text{bd}}}{\partial c_H} \right) + o(\eta).$$

If $\frac{\partial \mathcal{E}_{\text{bd}}}{\partial c_H}$ is larger than $\frac{\partial \mathcal{E}_{\text{bd}}}{\partial c_M}$, then the coefficient of η is negative. For sufficiently small positive η , the first-order term dominates the $o(\eta)$ remainder, so $\Delta \mathcal{E}_{\text{bd}} < 0$. The boundary error decreases while the total compression remains C . $\square$

B Algorithmic Summary

Algorithm 1 summarizes MedCore. The algorithm separates the scoring stage from the recovery stage. The reset-to-SAM operation is used only to compute importance. The final compressed model removes groups or keeps them active. It does not reset kept groups to SAM weights.

Algorithm 1 MedCore pruning

Require: MedSAM checkpoint θ^M , SAM checkpoint θ^S , frozen prompt encoder ϕ , calibration sets

$\{\mathcal{D}_{\text{cal}}^{(r)}\}_{r=1}^R$, structured groups $\mathcal{G}$, target budget B

Ensure: Pruned and recovered MedSAM checkpoint

- 1: Initialize all group masks as active.
 - 2: **for** each sub-distribution r **do**
 - 3: Compute boundary-aware MedSAM Fisher $F_{r,i}^M$ using Eq. (10).
 - 4: **end for**
 - 5: Compute SAM Fisher F_i^S on the same calibration data.
 - 6: **for** each group $g \in \mathcal{G}$ **do**
 - 7: Estimate $\hat{\Delta}_g^{\text{zero},(r)}$ using $F_{r,i}^M$.
 - 8: Estimate $\hat{\Delta}_g^{\text{reset},(r)}$ using Eq. (12).
 - 9: Compute the fused score $\hat{Q}_g^{(r)}$ and the distribution-aware score $\hat{Q}_g^{\text{dist}}$.
 - 10: Compute the cost-normalized priority P_g using Eq. (8).
 - 11: **end for**
 - 12: Allocate head-pruning budgets across blocks with block sensitivity.
 - 13: Prune attention heads with the lowest priorities under the block budgets.
 - 14: Run a short recovery fine-tuning stage.
 - 15: Recompute or reuse priorities for MLP connection groups in the head-pruned model.
 - 16: Prune MLP connection groups until the target budget is reached.
 - 17: Run post-pruning recovery fine-tuning with Eq. (14).
 - 18: Physically remove pruned structures and return the compact checkpoint.
-

C Additional Boundary Leverage Details

We compute boundary leverage on 384 calibration images, with 128 images from each polyp dataset. The sweep has seven head sparsity levels and seven MLP sparsity levels. We compute adjacent head steps and adjacent MLP steps. Steps with zero additional parameter reduction are undefined and are omitted.

For a head step, we fix the MLP sparsity and increase head sparsity. For an MLP step, we fix head sparsity and increase MLP sparsity. We use the same parameter-reduction unit for both families. This normalization is important because a 10% increase in head sparsity and a 10% increase in MLP sparsity do not remove the same number of parameters.

The boundary band width is tested at $\tau = 3$ and $\tau = 5$ pixels. Both choices give the same qualitative conclusion in our experiments. The main text reports the 95th-percentile boundary leverage because HD95 is also a high-percentile boundary metric. The mean boundary leverage gives the same direction but is less aligned with HD95.

The summary values are: $\text{BLR}_{95} = 2.887$, $\text{RLR} = 3.064$, $\text{BSR} = 0.942$, paired win rate = 1.000, and median paired difference = 2.603. BSR is the ratio between boundary leverage ratio and region leverage ratio. Since BSR is slightly below one, we do not claim that head pruning is boundary-specific. We claim that head pruning is high-boundary-leverage. This is the statement needed to explain boundary fragility.

D Implementation Details

We use the official MedSAM ViT-B checkpoint as θ^M and the SAM ViT-B checkpoint as θ^S . The prompt encoder remains frozen. Fisher estimation uses calibration images and the same bounding-box prompt protocol as MedSAM. We compute boundary maps by morphological dilation minus erosion of the binary mask. We normalize losses so that the boundary-weighted BCE term does not dominate the Dice term at the beginning of recovery.

For attention-head pruning, each group contains the corresponding Q, K, V, and output-projection parameters associated with one head. For MLP pruning, each group contains a connection group in the feed-forward block. We use physical removal after mask selection so that parameter count and FLOPs reflect the actual compact model.

The main one-time configurations protect the deepest encoder blocks because they have high Fisher sensitivity and connect directly to the mask decoder. The sequential extreme setting is different. It starts from the h70_m95 checkpoint and then performs a conservative second pruning pass on the previously protected deepest blocks. This setting is used only to test the limit of compression.

E Limitations

MedCore uses calibration data to estimate Fisher information and boundary leverage. If this calibration set is small or biased, the score may under-protect rare structures. The reset-to-SAM score also assumes that the SAM and MedSAM checkpoints are aligned parameter by parameter. This is true for the standard MedSAM adaptation path, but it may not hold for methods that change the architecture.

The boundary leverage theorem is local. It explains first-order boundary displacement under small logit perturbations. Very aggressive pruning can create non-local effects, such as disconnected masks or missing components, that are not fully captured by the first-order formula. We therefore view boundary leverage as a useful diagnostic and design principle, not as a complete replacement for final BF1 and HD95 evaluation.

Finally, our main experiments focus on 2D prompt-driven segmentation with bounding boxes. Extension to 3D medical segmentation and other prompt types requires additional validation. The same concepts can be applied, but the boundary band, distance metric, and computational cost must be adjusted for 3D volumes.